How the Wald interval is computed

V_{mech}, the covariance due to the DP noise, by parametric bootstrap at R: free post-processing

1 Simulate the noise

For b = 1, ..., B, around R's own release:

$$\mathbf{E}_b^* = \mathbf{E}_b^{*\top}, \quad E_{b,ij}^* \sim \mathcal{N}(0, \sigma_2^2) \text{ on } B, C;\ 0 \text{ on } A$$

$$\mathbf{f}_b^* \sim \mathcal{N}(0,\ \sigma_2^2 I_p)$$

$$\theta_b^* = \left(\hat{G}_2 + \mathbf{E}_b^* + \lambda\Psi\right)^{-1}\left(\hat{c}_2 + \mathbf{f}_b^*\right)$$

$$V_{\text{mech}} = \frac{1}{B-1}\sum_{b=1}^{B}\left(\theta_b^* - \bar{\theta}^*\right)\left(\theta_b^* - \bar{\theta}^*\right)^\top$$

keep b only if $\left\|\tilde{P}^{1/2}\,\mathbf{E}_b^*\,\tilde{P}^{1/2}\right\|_{\text{op}} \le \frac{1}{2}$

Without this truncation the bootstrap mean and covariance do not exist. Use antithetic pairs (E^*_b, f^*_b) and ($-E^*_b$, $-f^*_b$).

2 What it estimates

First order in the noise:

$$\tilde{\theta} - \theta_\lambda \ \approx\ \tilde{P}\left(\mathbf{f} - \mathbf{E}\,\theta_\lambda\right), \qquad \tilde{P} = \left(\hat{G}_2 + \lambda\Psi\right)^{-1}$$

$$V_{\text{mech}} \ \approx\ \sigma_2^2\ \tilde{P}\left(I + K(\tilde{\theta})\right)\tilde{P}$$

$$K(\theta)_{ii} = \sum\nolimits_j \chi_{ij}\,\theta_j^2, \qquad K(\theta)_{ik} = \chi_{ik}\,\theta_i\theta_k \quad (i \ne k)$$

χ_{ij} = 0 if i and j are both in R's block A (never noised), else 1.

I: noise f on $\hat{c}_2$. K(θ): noise E on $\hat{G}_2$, times θ.

R can use this closed form instead of the bootstrap: same first-order guarantee, no Monte Carlo error.

3 The interval

$$\tilde{\theta}_j \ \pm\ z_{1-\alpha_{\text{CI}}/2}\sqrt{(V_{\text{mech}})_{jj}}$$

$\alpha_{CI} = \alpha - \alpha_L - \alpha_U = 0.045$, so z = 2.005 (nominal 95.5%)

Target

$$\theta_\lambda = (G + \lambda\Psi)^{-1}\,c$$

a fixed function of the data, so only the noise enters: no sampling variance.

First order: relative error $O(1/\vartheta_0) + O(B^{-1/2})$. In simulation V_{mech} is within 2% of the true noise s.d., and the intervals cover 0.944–0.962 (target 0.955). The interval for β adds sampling variance, $V = V_{mech} + \tilde{s}^2\,\tilde{P}\,\hat{G}_2\,\tilde{P}$, and is widened for the ridge bias.